EXPERIMENT 10

A* Algorithm

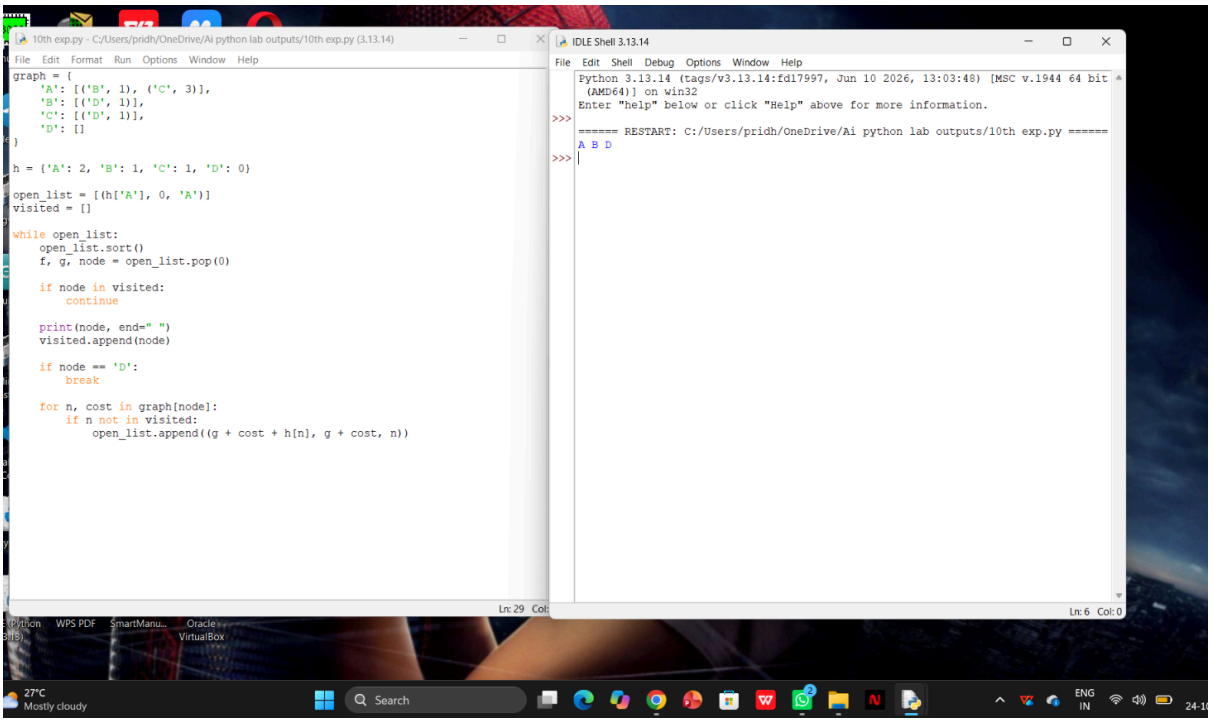
Aim:

To write a Python program to implement the A* search algorithm to find the shortest path between two nodes.

Algorithm:

- Initialize the starting node and goal node.
- Calculate the evaluation function
- f(n)=g(n)+h(n).
- Store the nodes to be explored in a priority queue.
- Select the node with the lowest evaluation value.
- Update the costs of its neighboring nodes.
- Continue until the goal is reached and display the path.

Program output :



Result: Thus, the Python program to implement the A* search algorithm was executed successfully.